

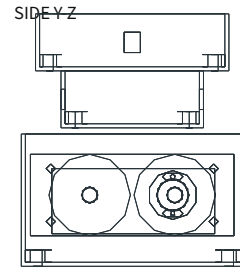
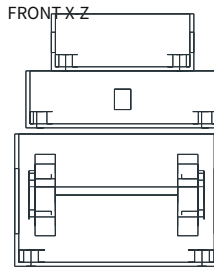
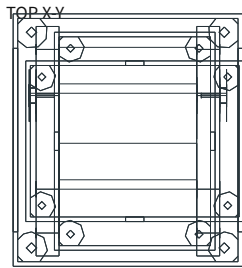
1 FM-002 — gauge/puller

물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

FM-002 gauge puller

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 11



Overall X 100.50 · Y 110.00 · Z 124.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: 6061 plates / silicone rollers / S45C eccentric bushes
- 핵심 치수/공차: adjusted unloaded roller gap 1.60–1.90 mm over full rotation; bush index pair $\leq 0.5^\circ$; roller axes parallel $\leq 0.05/80$; gauge datum alignment ≤ 0.10 mm
- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: d09fe9bc0a587c2aadf27557c7d2bed8bf3f9424 · 물리 검증: NOT_RUN